

Release_Change_Management_Standard

Technical Standard

保险公司 技术开发部
内部文档 · 仅供授权人员查阅

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Release & Change Management Standard

> Reference: ITIL Change Management Best Practices, Tencent Change Management Standards, Google SRE Change Management

1. Overview

1.1 Purpose

Standardize the change management process within the Technology Development Department, ensuring all changes are classified, reviewed, approved, and executed in a controlled manner, reducing the risk of production incidents.

1.2 Scope

Applies to all changes affecting the production environment, including but not limited to: application releases, database changes, configuration modifications, infrastructure changes, and third-party service switches.

1.3 Terminology

Term	Definition
Change	Any modification to production or staging environments
CAB	Change Advisory Board
Change Window	Fixed time period for executing high-risk changes (blackout window)
RTO	Recovery Time Objective
RPO	Recovery Point Objective

2. Change Classification

2.1 Change Categories

Category	Definition	Example
Standard	Pre-authorized, low-risk, fixed process	Routine config changes, IP whitelist
Normal	Requires review and approval	Feature release, DDL changes
Emergency	Quick execution for incident fix	P0/P1 hotfix, security patch
Major	High-risk, cross-team, broad impact	Architecture refactoring, core system migration

2.2 Change Level Matrix

Level	Impact	Risk	Approval Chain	Window
L1 (Low)	Single service/module	Low	Tech Lead	Business hours
L2 (Medium)	Multiple services/modules	Medium	Tech Lead + Architect	Pre-release window
L3 (High)	Cross-system, business impact	High	CAB review	Change window
L4 (Major)	Platform-wide, core system	Critical	CTO/VP + CAB	Special window

2.3 Standard Change List

Pre-authorized standard changes (record required, no individual approval):

- Monitoring alert threshold adjustments
- IP whitelist additions/removals
- Log level adjustments (non-production)
- Non-functional config toggles in config center
- Read-only user authorization

3. Approval Process

3.1 Normal Change Process

Request > Impact Assessment > Technical Review > Approval > Schedule > Execute > Verify > Close

Step	Owner	Output	Timeline
1. Request	Initiator	Change ticket	---
2. Impact Assessment	Tech Lead	Impact analysis	2h
3. Technical Review	Architect (L2+)	Review comments	4h
4. Approval	Approver (per level)	Decision	4h
5. Schedule	Tech Lead	Execution schedule	---
6. Execute	Developer/Ops	Execution log	Per schedule
7. Verify	Initiator + QA	Verification report	30min post-execution
8. Close	Tech Lead	Ticket closed	After verification

3.2 Emergency Change Process

Incident > Emergency Assessment > Emergency Approval (post-execution OK) > Execute > Verify > Post-hoc Ticket

- Emergency approvers: Tech Lead + Architect + On-call DBA
- Post-hoc ticket and postmortem required within 24 hours
- If a service has 3+ emergency changes in a month, root cause analysis required

3.3 Major Change Process

- Submit change plan 1 week in advance
- CAB meeting required for review
- Full-link load testing and canary required
- Detailed rollback plan and emergency response required
- Notify all affected stakeholders in advance
- At least 2 on-call personnel during execution

4. Change Window

4.1 Definition

A change window is a fixed time period during which high-risk changes may be executed, requiring key stakeholders to be online.

4.2 Schedule

Item	Description
Regular window	Every Thursday 22:00 - 02:00
Emergency window	On-demand, CTO approval required
Cooldown period	Last 3 days of month + 3 days before holidays (emergency only)
Annual freeze	2 weeks before Spring Festival to 1 week after (P0 fix only)

4.3 Rules

- Changes must be approved 2 hours before the window
- Changes not started within 30 minutes of window opening are auto-postponed
- Unfinished changes at window close: rollback if in reversible state
- No parallel L3 changes during the same window

5. Change Ticket Template

```
# Change Ticket - [CHANGE-YYYYMMDD-001]
## Basic Info
- Title: [Brief description]
- Related Requirement: [JIRA-ID / BRD-ID]
- Category: [Standard/Normal/Emergency/Major]
- Level: [L1/L2/L3/L4]
- Initiator: [Name]
- Tech Lead: [Name]
## Description
### Scope
[Affected systems, modules, services]
### Content
[Detailed technical description]
### Expected Impact
[Business impact description]
## Solution
### Steps
1. [Step 1]
2. [Step 2]
### Rollback Plan
[Detailed rollback steps]
### Canary Strategy
[Ratio, verification metrics, promote conditions]
## Risk Assessment
| Risk | Probability | Impact | Mitigation |
| --- | --- | --- | --- |
| [Risk 1] | L/M/H | L/M/H | [Action] |
## Approval History
| Reviewer | Decision | Time | Signature |
| --- | --- | --- | --- |
| [Name] | [Approve/Reject] | [Time] | --- |
## Execution Log
| Step | Time | Executor | Result |
| --- | --- | --- | --- |
| [Step 1] | [Time] | [Name] | [Success/Fail] |
## Verification
```

Item	Result	Verifier
---	---	---
[Item 1]	[Pass/Fail]	[Name]

6. Change Review

6.1 Weekly Summary

Weekly change review meeting covering:

- Total changes for the week
- Count by level
- Success rate
- Rollbacks and reasons
- Emergency change count and root causes

6.2 Metrics

Metric	Target	Alert Threshold
Change success rate	> 99%	< 98%
Emergency change ratio	< 10%	> 20%
Rollback rate	< 5%	> 10%
Change-induced incidents	0	1
Average lead time	< 3 days	> 7 days

7. Appendix

7.1 Reference Documents

- ITIL 4 Change Management Practices
- *CICD Standard*
- *Release Checklist Standard*
- *Emergency Response Plan & SOP*

7.2 CAB Composition

Role	Member	Responsibility
Chair	Tech Director	Lead review, final decision
Architect	Core architect	Review technical solutions
Ops Representative	Ops lead	Review infrastructure impact
Security Representative	Security lead	Review security impact
Business Representative	Business lead	Assess business impact
Change Manager	PM	Process management and records